

CUSTOMER MILESTONE BRIEF

MS #19 - DB2 and Multi-Target Data Modernization

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.19.0, 0.19.1, 0.19.2
Date	2026-08-22 to 2026-08-23
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #19 - DB2 and Multi-Target Data Modernization converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Introduces a database modernization proof that can describe DB2 structures and compare more than one target without hard-coding a single destination.

Why this matters to a customer

Introduces a database modernization proof that can describe DB2 structures and compare more than one target without hard-coding a single destination.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added deterministic Db2 DDL, DCL, and embedded-SQL parsing for the CardDemo AUTHFRDS vertical slice.
- Added first-class Db2 table, column, constraint, index, DCL, and SQL-statement entities to the evidence graph.
- Added paragraph-to-SQL, SQL-to-table, and SQL-to-column lineage plus four curated data business rules.
- Added a target-neutral canonical model, PostgreSQL 16 adapter, schema, migration mapping, and boundary fixtures.
- Added fail-closed schema/data/query/transaction equivalence checks and a tamper-evident signed development receipt.
- Added an isolated Docker PostgreSQL proof command and customer-key receipt signing path.
- Added a Control Tower Data panel for lineage posture, equivalence checks, and unresolved production gaps.

- Added 17 focused tests, including empty-input, duplicate-key, incomplete-row, missing-marker, and signature-tamper cases.
- Wait for the requested PostgreSQL database to accept a real query before running the live equivalence proof.
- Record bounded live-proof failure reason and psql exit code without persisting raw database output.
- Add regression coverage for the PostgreSQL entrypoint initialization race.
- Generalized data-target generation and catalog expectations behind a versioned adapter contract.
- Added an Oracle Database 26ai Free adapter, schema projection, boundary-fixture loader, Docker runner, and explicit Oracle semantic gaps.
- Replaced coarse PostgreSQL object counts with exact column, type, length, precision, scale, nullability, key-order, and index-order evidence.
- Added normalized row-level checksums, bounded query comparisons, and independent commit/rollback probes for both targets.
- Added fail-closed parsing for missing, malformed, unknown, and duplicate evidence markers plus adversarial mutation tests.
- Bound model, mapping, fixture, schema SQL, fixture SQL, verification SQL, adapter version, and container image identity into live receipts.
- Added a multi-target aggregate receipt that cannot pass unless both PostgreSQL and Oracle receipts pass.
- Added a side-by-side target evidence matrix to the Control Tower while retaining `production_ready: false`.
- Corrected Oracle SQL/JSON generation after the first live 26ai Free probe: apply NULL ON NULL once per object, emit bounded one-line VARCHAR2 evidence, and exit SQL*Plus on the first database error.

How it differs from earlier work

MS #19 builds on MS #18 (CICS, VSAM, IMS, and HLASM Readiness Proofs). The earlier milestone established its own bounded capability; this milestone adds db2 and multi-target data modernization while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use db2 and multi-target data modernization as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- This milestone establishes only the bounded capabilities and evidence listed here; later capabilities are not retroactive.
- Production readiness and platform equivalence require the explicit evidence and policy gates applicable to the customer environment.

Source of record

- CHANGELOG.md release section(s): 0.19.0, 0.19.1, 0.19.2
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity